

Orbital Mass and Proptosis

Guidelines for Clinical Case Presentations

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HISTORY:

Presenting Complaints:

- Proptosis
- Pain
- Loss of Vision
- Diplopia

Details:

Proptosis (“Prominent eyeball”)

- Duration and age of onset
- Unilateral or bilateral
- Onset: Insidious or Sudden
- Progression or Stationary or Regression
- Constant or Intermittent
- Variation of proptosis: Postural variation or Cough or Strain
- Aggravating or Relieving factors
- Associated “swishing sound” (carotid-cavernous sinus fistula)

Pain

- Duration and relation to the onset of proptosis
- Site
- Radiation
- Character
- Aggravating and relieving factors
- Relation to eye movement

Vision loss

- Duration and relation to the onset of proptosis
- Onset: Insidious or Sudden
- Progression: Rapid or Slow
- Gaze evoked amaurosis

Diplopia

- Duration and relation to the onset of proptosis
- Monocular or Binocular
- Horizontal or Vertical
- Variability
- Maximum field of diplopia

Other Relevant History

- Ptosis
- Lid retraction and edema
- Redness, watering and photophobia (especially with exposure keratopathy)
- Infraorbital tingling or numbness
- Swelling anywhere else on the body
- Trauma
- CNS: Headache, nausea vomiting, convulsions, ataxia
- Focal sepsis:
 - “Boil” on the eye lid or face
 - Dacryocystitis: Swelling near medial canthus, discharge, matting of lashes
 - Sinusitis: Headache (frontal), nasal stuffiness or discharge, fever
 - Dental sepsis: Tooth pain
 - Ear: Discharge, tinnitus or hearing loss

Inflammatory or neoplastic lesions present with an **acute onset**. Benign lesions usually have a **chronic onset**.

Exophthalmos occurs in 30% of the patients suffering from an orbital tumor. However, pain, diplopia, swelling, tearing and blurred vision are much more common presenting complaints. Since most optic nerve gliomas occur in the first two decades of life, loss of vision in children with or without optic atrophy suggests optic nerve glioma, particularly if some degree of exophthalmos exists. In a child, the presence of a retrobulbar mass with ecchymosis of the lids and subconjunctival hemorrhage associated with exophthalmos suggests a malignant tumor, particularly medulloblastoma.

Past History:

- Diabetes or hypertension
- Medical history
- Ocular history
- Previous surgeries
(Mastectomy, Thyroid, Nasal, Trauma)

Mucormycosis should be considered in a **diabetic patient** suffering from a rapidly advancing exophthalmos with loss of vision and impaired motility.

Personal History:

- Weight loss

Family History

Systemic Review:

- CVS
- RS
- GIT
- Renal
- Musculoskeletal

EXAMINATION:

General Physical Examination

- General physical appearance
- Vitals: Pulse, BP, Respiratory Rate
- Pallor, Icterus, Cyanosis, Edema
- Lymph nodes
- Thyroid
- Skin: Naevi, Café-au-lait, Port wine stain

Order of Relative Frequency of Causes of Exophthalmos:

- Thyroid Disease
- Inflammation
- Orbital Tumors
- Vascular Disorder
- Orbital Anomaly

Systemic Examination

- Ear, Nose and Throat
- CNS
- CVS
- RS
- Abdomen: specifically look for an abdominal mass and organomegaly
- Musculoskeletal

OCULAR EXAMINATION

Overview

Simple Inspection is the first step in any physical diagnosis. The most intense rubor (redness) occurs with orbital cellulitis. Orbital and lid inflammation and minimal exophthalmos may occur after trauma to a dermoid tumor that has caused extrusion of its contents. Inflammation can also accompany rapidly growing tumors such as rhabdomyosarcoma in children.

Most orbital tumors, particularly those which are benign, occur in the upper two quadrants and displace the eye downward (eccentric displacement as opposed to an axial displacement of the globe). If the eye is displaced upward, consider lymphoma or carcinoma extending from the maxillary sinus. Lateral displacement occurs with mucocele or malignant tumors from the ethmoid sinuses. Posterior orbital tumors cause more exophthalmos than displacement. An exception is the hemangioma of infancy which tends to wrap around the eye rather than displace it.

Since most of the benign tumors are located anteriorly, palpation by the small finger will frequently uncover a mass. These include lacrimal gland, dermoid cyst, hemangioma and lymphoma.

Suggested Order of Ocular Examination:

1. General Ocular Examination
2. Examination of Proptosis
3. Detailed Ocular Examination
(including fundus, refraction and intraocular pressure and ocular motility)

General Ocular Inspection

- Head Posture
- Ocular Posture
- Facial asymmetry

Examination of Proptosis

Inspection

- Unilateral or Bilateral
- Axial or Eccentric
- Visible pulsatility
- Change with Valsalva, cough or posture
- Lid position: Retraction or ptosis
- Lid fullness
- Skin over the lid
 - Inflammation
 - Pigmentation (Hyper or Hypo)
 - Ecchymosis
 - Venous dilatation

Watch for pseudoproptosis:

Lid retraction can occur without exophthalmos in thyroid disease. A small ptosis in the other eye may give the appearance that the other, normal, lid is retracted. The patient complains about the cosmetic appearance of the eye that bothers him the most when in reality the problem may be in the other eye.

Palpation

- Rise in temperature
- Tenderness
- Pulsatility
- Thrill
- Mass palpated
 - Size and shape
 - Discrete or ill-defined
 - Margins
 - Consistency
 - Compressibility
 - Ability to insinuate a finger between the mass and bony orbit
- Resistance to retropulsion of the globe
- Auscultation over the orbit

An arteriovenous fistula commonly causes **pulsating exophthalmos** in an adult. In children, pulsating exophthalmos is usually caused by neurofibromatosis with a dehiscence in the orbital roof. Subtle pulsations may be observed only when looking at the fundus with a direct ophthalmoscope or when performing applanation tonometry.

A **bruit** is better heard with the bell of a stethoscope.

Exophthalmometry

- Axial: from the lateral orbital rim to the tip of the cornea
 - Normal: 16 to 18 mm (Difference of 2mm)
 - Minimal: 20 to 23 mm (Difference of 3 to 4mm)
 - Moderate: 24 to 27 mm (Difference of 5 to 7mm)
 - Severe: More than 28 mm (Difference of more than 8mm)
- Vertical
- Horizontal

The measurements of a displaced globe may be expressed in absolute, comparative and relative terms.

Measuring Exophthalmos:

It is preferable to use the Hertel instrument as it provides better reproducibility. It measures the relationship between the lateral orbital rim and the anterior cornea surface. This instrument allows the examiner to reproduce the same pupillary distance from exam to exam. It also allows the examiner to keep the same parallax view for observation of the corneal surface by lining up two lines in its mirrors.

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Detailed Ocular Examination

- Visual Acuity
 - Acuity (Distance and Near)
 - Colour vision
 - Confrontation field of vision
- Eyelid (included with inspection of proptosis)
 - Palpebral fissure height
 - Lagophthalmos: Lid closure, blink rate
 - Eye lid sensation (include forehead and cheek)
 - Bells phenomenon
 - Special: Lid signs of thyroid ophthalmopathy
- Conjunctiva
 - Chemosis, Congestion, Haemorrhage
 - Conjunctival extension of orbital tumors (eg. Salmon patch)
- Cornea
 - Size, clarity
 - Keratic Precipitates
 - Staining
 - Sensation
- Anterior Chamber
 - Depth
 - Reaction (Cells and Flare)
- Iris
 - Colour, Pattern, Atrophy, Neovascularisation
- Pupil: afferent pupillary defect, tonic pupil (light near dissociation)
- Lens
- Fundus: Distant Direct, Direct and Indirect Ophthalmoscopy
 - Disc Oedema, Optic Atrophy
 - Posterior Pole Indentation, Choroidal Folds, Optociliary shunts
 - Tortuous vessels
- Refraction
- Ocular Motility: Ductions and Versions
- Intraocular Pressure
 - Straight ahead gaze
 - In Upgaze for Thyroid Ophthalmopathy

Visual acuity loss can occur with exophthalmos from any cause; however, a retrobulbar tumor, particularly in the muscle cone, is the first consideration in the case of slowly progressive exophthalmos associated with loss of acuity. If the acuity can be restored by plus lenses and if increasing hyperopia is suggested, a retrobulbar tumor that is causing a flattening of the posterior surface of the eye should be considered. This is not pathognomonic of a retrobulbar tumor, since it can be seen also in posterior scleritis or with mild subretinal edema from other causes displacing the retina anteriorly.

In cases of endocrine exophthalmos, loss of vision is usually minimal and is frequently variable, suggesting some problem with the corneal epithelium secondary to exposure or to disturbance in the tear film from dryness or decreased blinking. In some cases, when exophthalmos progression is rapid and accompanied by considerable chemosis and inflammation, the loss of vision can be profound and may be caused by optic nerve involvement.

Tumors that indent the posterior part of the globe, particularly in the muscle cone, can cause **horizontal striae in the macula**. The folds are seen not only with retrobulbar tumors, but may also be seen in association with ocular hypotony, intraocular tumors, scleritis, uveitis, after scleral surgery, and occasionally in thyroid disease.

Special Examination

- Diplopia Charting (if diplopia is present)
- Forced Duction Test (thyroid ophthalmopathy and sequelae of orbital trauma)

Evaluation of Motility

Orbital tumors and some endocrine disorders can cause exophthalmos without impairment of motility. Benign orbital masses frequently cause only minor limitation of local rotation, usually in the field of action where the mass is located. Although the two conditions usually occur together, in some cases of endocrine disease ophthalmoplegia may occur before frank exophthalmos appears. Forced duction tests are usually positive in endocrine disease. However, some limitation may be experienced - and the test is less than specific - if considerable orbital edema or inflammation from other causes exists. Endocrine ophthalmoplegia usually manifests as limitation of upgaze, from restriction of the inferior rectus and inferior oblique in the area of Lockwood's ligament.

Laboratory Studies

Such studies are usually of little value. If leukemia or multiple myeloma is suspected, a complete blood count is revealing. In cases of lymphoma an evaluation of the bone marrow is helpful. Thyroid studies are often disappointing because thyroid exophthalmos and ophthalmoplegia may occur when the thyroid appears to be functioning normally at least as indicated by blood tests.

Radiographic Studies

The best test for thyroid disease is a CT or MRI of the orbit. Either one will demonstrate large muscles even in the absence of diplopia. However, you must order coronal views or you will not get views of the vertical muscles except when you are lucky. This test is particularly useful in cases where there is minimal exophthalmos but loss of vision is the main problem. It will demonstrate the enlarged muscles around the optic nerve posteriorly. MRI is the better choice, however, for looking at other orbital soft tissue entities.